

5 V 1 A Flyback Transformer — Final Characterization

(TinySwitch Series)

Executive Summary

The transformer constructed is suitable for a 5 V @ 1 A TinySwitch-III/IV flyback converter operating from 120 VAC. Measured magnetizing inductance, turns ratio, and leakage inductance align closely with design targets for a ~130 kHz converter. The final assembly was rigid, well-gapped, and achieved strong coupling with adequate drain-voltage margin after snubbing.

Final Measured Parameters (100 kHz, fully assembled)

Parameter	Measured Value	Notes
Primary magnetizing inductance (Lm)	2.476 mH	Pins 1–3, 119 T
Secondary inductance (Ls)	26.64 μ H	Secondary pins, short leads
Auxiliary inductance (Laux)	26.77 μ H	Aux pins, short leads

Derived Quantities

Turns ratio from inductance: $N_p/N_s \approx \sqrt{L_p/L_s} = \sqrt{(2476 \mu\text{H} / 26.64 \mu\text{H})} \approx 9.64$. With $N_p = 119$, inferred $N_s \approx 12$ turns (aux ≈ 12 T). Reflected voltage for 5 V output (Schottky ≈ 0.5 V): $V_{OR} = (N_p/N_s)(V_o + V_d) \approx 9.64 \times 5.5 \approx 53$ V. Estimated drain peak: $V_{D,pk} \approx 185 \text{ V} + 53 \text{ V} + \text{spike} \leq 280 \text{ V}$ after snubber.

Leakage Inductance Characterization (Primary measured; others shorted as noted)

Shorting Configuration	L_leak @ 10 kHz	L_leak @ 100 kHz	% Leakage	Coupling (k)
Secondary only shorted	59.45 μ H	53.10 μ H	2.14%	0.989
Secondary + Aux shorted	53.77 μ H	46.58 μ H	1.88%	0.991

Use the conservative leakage (53.1 μ H) for RCD snubber sizing. “All-secondaries-shorted” readings confirm excellent coupling (~ 0.991).

RCD Snubber — First-Cut Values

Assume $I_{pk} \approx 0.20 \text{ A}$ @ 5 W, $\eta \approx 0.8$, $f \approx 132 \text{ kHz}$.

Case	L_leak	E_leak/cycle	P_leak @ 132 kHz
Conservative (Sec shorted, 100 kHz)	53.10 μ H	1.062 μ J	140.2 mW
Worst @ low f (Sec shorted, 10 kHz)	59.45 μ H	1.189 μ J	156.9 mW
Best case (Sec + Aux shorted, 100 kHz)	46.58 μ H	0.932 μ J	123.0 mW

Design starting point: • $C_{snub} = 1.5\text{--}2.2 \text{ nF}$ (C0G/NP0 dielectric) • $D_{snub} \geq 600\text{--}1000 \text{ V}$, fast recovery • $R_{snub} = 150\text{--}330 \text{ k}\Omega$, $\geq 0.5 \text{ W}$ Tune until $V_{D,pk} \leq 280 \text{ V}$ and ringing $\zeta \approx 0.5\text{--}0.7$.

Verification & Bring-Up Checklist

• Primary DCR (pins 1–3), R_{series} @ 100 Hz $\approx 4.5 \Omega$ • Confirm turns ratio (1 Vrms @ 1 kHz): $V_p/V_s \approx 1/9.64$; mark dot polarity • Low-voltage test (20–30 V DC) with light load: verify 5 V regulation and auxiliary behavior • Scope drain (diff probe): tune RCD for ≤ 280 V peaks and moderate damping • Thermal test & 1 A load test; then proceed to AC input via isolated variac

Prepared for: SMPS-5V1A Project

Note: Data reflect rigidly clamped transformer measured at 100 kHz unless otherwise noted.